

Bayesian Inference of Genetic Information for Disease Discrimination

Pólya-gamma Gibbs Sampler Targeting the lr_discrete_blockdiag Model

March 2026

1 Introduction

A central quantity in genetic epidemiology is the expected log-likelihood ratio (weight of evidence, in nats) for predicting an individual's disease status from their genome, known here as **genetic information** μ . Under a Gaussian liability model this equals $0.5s^2$, where s is the scale of the class-conditional log-likelihood ratio distribution. Equivalently, $\lambda_S = e^\mu$ is the sibling recurrence-risk ratio, the standard measure of familial aggregation.

The goal is to infer μ from a case-control dataset in which some sampled individuals are biological relatives, without access to individual genotypes. Instead, the genetic relationship matrix (GRM) Φ , computed from genome-wide SNP data, is used to describe the covariance structure.

2 Statistical Model

2.1 Generative model

Let $\mathbf{y} \in \{0, 1\}^M$ denote case-control status for M individuals and let $\Phi = LL^\top$ be the Cholesky decomposition of the GRM. The model is a logistic regression with a two-component mixture random effect:

$$\mu \sim \text{HalfCauchy}(2), \quad \phi = \text{logit}(p) \sim \mathcal{N}(0, \sigma_\phi^2), \quad p \sim \text{Beta}(20 p_{\text{obs}}, 20(1 - p_{\text{obs}}))$$

For each individual $i = 1, \dots, M$:

$$\begin{aligned} r_i \mid p &\sim 2 \text{Bernoulli}(p) - 1 \in \{-1, +1\}, \\ Z_{\text{mix}, i} \mid r_i, \mu &\sim \mathcal{N}(r_i \mu, 2\mu), \\ \eta_i &= \phi + (L\mathbf{Z}_{\text{mix}})_i - \bar{\Delta} (L\mathbf{1})_i, \quad \bar{\Delta} = (2p - 1) \mu, \\ y_i \mid \eta_i &\sim \text{Bernoulli}(\sigma(\eta_i)), \quad \sigma(x) = (1 + e^{-x})^{-1}. \end{aligned}$$

The binary indicator r_i determines the class (+1 = genetic risk carrier, -1 = non-carrier); the mixing proportion p estimates the population prevalence K . The Cholesky factor L is block-diagonal, one block per connected family cluster. The shift $\bar{\Delta}(L\mathbf{1})_i$ centres the logit on the marginal mean of Z_{mix} .

The theoretical constraint $\mu = \frac{1}{2}s^2$ (variance equals twice the mean for the genotypic value) is enforced through the parameterisation $Z_{\text{mix},i} \mid r_i, \mu \sim \mathcal{N}(r_i\mu, 2\mu)$, so the marginal genotypic-value distribution is

$$(L\mathbf{Z}_{\text{mix}})_i \sim \frac{1}{2}\mathcal{N}(\mu, 2\mu) + \frac{1}{2}\mathcal{N}(-\mu, 2\mu).$$

2.2 Reduction to relatives

Unrelated individuals (singletons in the GRM) contribute no information about λ_S : their Z_{mix} terms do not covary with any other individual, so after marginalising Z_{mix} the likelihood factors for singletons are flat in μ . All singletons are therefore discarded before fitting, retaining only the $M_{\text{rel}} \ll M$ individuals in family blocks of size ≥ 2 .

3 Pólya-gamma Gibbs Sampler

3.1 Pólya-gamma augmentation

Inference is performed by Gibbs sampling under the Pólya-gamma (PG) augmentation of Polson, Scott and Windle (2013). Introducing auxiliary variables $\omega_i \sim \text{PG}(1, |\eta_i|)$, the logistic likelihood becomes conditionally Gaussian:

$$\kappa_i \mid \omega_i, \eta_i \sim \mathcal{N}\left(\omega_i \eta_i, \omega_i^{-1}\right), \quad \kappa_i = y_i - \frac{1}{2}.$$

Given $\boldsymbol{\omega}$, the full conditional of $\mathbf{Z}_{\text{mix}}$ is Gaussian with precision matrix $L^\top \text{diag}(\boldsymbol{\omega})L + \tau I$ (where $\tau = 1/(2\mu)$ from the Gaussian prior) and a linear right-hand side.

3.2 Full Gibbs sweep

The LRDiscreteBlockdiagPGGibbs sampler performs the following steps each iteration:

1. **Sample $\boldsymbol{\omega}$:** $\omega_i \mid \eta_i \sim \text{PG}(1, |\eta_i|)$, implemented via the `polygamma` C extension.
2. **Sample $\mathbf{Z}_{\text{mix}}$:** joint Gaussian draw for each family block b ,

$$\mathbf{Z}_{\text{mix},b} \mid \boldsymbol{\omega}_b, \kappa_b, \phi, \mu, \mathbf{r}_b \sim \mathcal{N}(A_b^{-1}\mathbf{b}_b, A_b^{-1}),$$

where $A_b = L_b^\top \text{diag}(\boldsymbol{\omega}_b)L_b + \tau I$ and $\mathbf{b}_b = L_b^\top(\boldsymbol{\kappa}_b - \boldsymbol{\omega}_b c_b) + \frac{1}{2}\mathbf{r}_b$, with $c_b = \phi - \bar{\Delta}v_b$ the per-individual offset and $v_b = (L_b\mathbf{1})_b$. For size-2 blocks the Cholesky solve is fully vectorised without LAPACK calls.

3. **Sample $\mathbf{r}$:** exact Bernoulli draw per individual,

$$p(r_i = +1 \mid Z_{\text{mix},i}, \phi) = \sigma(Z_{\text{mix},i} + \phi).$$

4. **Sample ϕ :** univariate slice sampler on the PG-augmented log-posterior,

$$\log p(\phi \mid \boldsymbol{\omega}, \mathbf{Z}_{\text{mix}}, \mathbf{r}, \mathbf{y}) \propto \boldsymbol{\kappa}^\top \boldsymbol{\eta}(\phi) - \frac{1}{2}\boldsymbol{\omega}^\top \boldsymbol{\eta}(\phi)^2 + (a-1)\log p + (b-1)\log(1-p) + n_+\log p + n_-\log(1-p),$$

where n_+, n_- are the current counts of $r_i = +1$ and $r_i = -1$, and $a = 20p_{\text{obs}}$, $b = 20(1 - p_{\text{obs}})$ are the Beta prior parameters.

5. **Sample μ (collapsed):** slice sampler on the log-posterior $p(\theta \mid \boldsymbol{\omega}, \mathbf{r}, \phi, \mathbf{y})$ with $\theta = \log \mu$, marginalising $\mathbf{Z}_{\text{mix}}$ analytically. For each family block b the eigendecomposition $A_b = U_b \text{diag}(\boldsymbol{\lambda}_b) U_b^\top$ yields

$$\log p(\theta \mid \dots) \propto \log p(\theta) - \frac{1}{2} M_{\text{rel}} \log(2\mu) - \frac{1}{4} M_{\text{rel}} \mu + C(\boldsymbol{\omega}, \phi, \mu) + \frac{1}{2} \sum_b \sum_j \frac{(p_{0,bj} + \mu \bar{m} p_{1,bj})^2}{\lambda_{bj} + \tau} - \frac{1}{2} \sum_b \sum_j \log(\lambda_{bj} + \tau)$$

where $\bar{m} = \tanh(\phi/2)$, $\tau = 1/(2\mu)$, $p_{0,bj} = (U_b^\top \mathbf{b}_{0,b})_j$, $p_{1,bj} = (U_b^\top \mathbf{b}_{1,b})_j$ are projections of the μ -independent and μ -dependent parts of the right-hand side onto the eigenvectors, and $C(\boldsymbol{\omega}, \phi, \mu)$ collects the offset-only quadratic terms.

6. **Refresh $\mathbf{Z}_{\text{mix}}$:** repeat step 2 under the updated μ to improve mixing.

3.3 Prior on μ

The half-Cauchy prior $\mu \sim \text{HalfCauchy}(2)$ is used throughout. In the $\theta = \log \mu$ parameterisation the log-prior (including the Jacobian) is

$$\log p(\theta) = \theta - \log \left[1 + \left(e^\theta / 2 \right)^2 \right] + \text{const},$$

which is flat around $\theta = 0$ ($\mu = 1$) and provides regularisation against extreme values.

4 Simulation Study

4.1 Simulated population and case-control sample

The generative model was simulated with true $\mu = 2.0$ ($\lambda_S = e^2 \approx 7.39$) and disease prevalence $K = 0.01$. The population comprised 400,000 full-sib pairs (genetic correlation 0.5), 200,000 full-sib triplets (genetic correlation 0.5), 40,000 half-sib pairs (genetic correlation 0.25), and 4,000 unrelated individuals (1,484,000 total). A 1:1 case-control sample retained all 14,761 cases and an equal number of controls.

Table 1: Case-control sample characteristics (seed = 42, true $\mu = 2.0$, $K = 0.01$).

Quantity	Count
Total sample M	29,522
Cases / Controls	14,761 / 14,761
Full-sib pairs (both sampled)	283
of which concordant-affected	166
Triplet sib-pairs (both sampled)	400
concordant case-case	241
discordant	107
concordant ctrl-ctrl	52
Half-sib pairs (both sampled)	19
M_{rel} (block size ≥ 2)	1,374
Size-2 blocks (pairs)	672
Size-3 blocks (triplets)	10

4.2 Comparison with DiscreteHMC Gibbs

The `LRDiscreteBlockdiagPGGibbs` sampler was compared with NumPyro’s `DiscreteHMC Gibbs` (NUTS inner kernel, `modified=True`) on the same data. Both algorithms operated on the same $M_{\text{rel}} = 1,374$ relatives. Four chains were run for each algorithm.

Table 2: Posterior summaries for μ (true value: 2.0). ESS: effective sample size (bulk); $\hat{R}$: Gelman-Rubin statistic.

Algorithm	Warmup	Samples	Median	SD	90% CI	ESS	$\hat{R}$	Wall time
PG-Gibbs	1,000	5,000	2.167	0.993	[1.253, 4.230]	189	1.030	87 s
DiscreteHMC Gibbs	2,000	2,000	2.134	0.889	[1.259, 3.937]	250	1.020	416 s

The posterior medians agree closely (2.167 vs 2.134) and both 90% credible intervals contain the true value $\mu = 2.0$. The PG-Gibbs sampler is approximately $5\times$ faster in wall time (87 s vs 416 s) on CPU versus GPU respectively, at a cost of lower ESS per sample due to slower mixing.

4.3 Discussion

Agreement between samplers. The close agreement in posterior medians (difference $< 2\%$) and overlapping credible intervals confirm that `LRDiscreteBlockdiagPGGibbs` is targeting the same posterior as `DiscreteHMC Gibbs`. The wider credible interval from PG-Gibbs (width 2.977 vs 2.678) reflects somewhat slower mixing (IAT ≈ 106 vs ≈ 32).

Information content. With $K = 0.01$ and 1:1 case-control matching, most relative pairs in the sample are discordant, carrying less information about λ_S than concordant-affected pairs. The half-Cauchy(2) prior retains considerable influence with $M_{\text{rel}} = 1,374$.

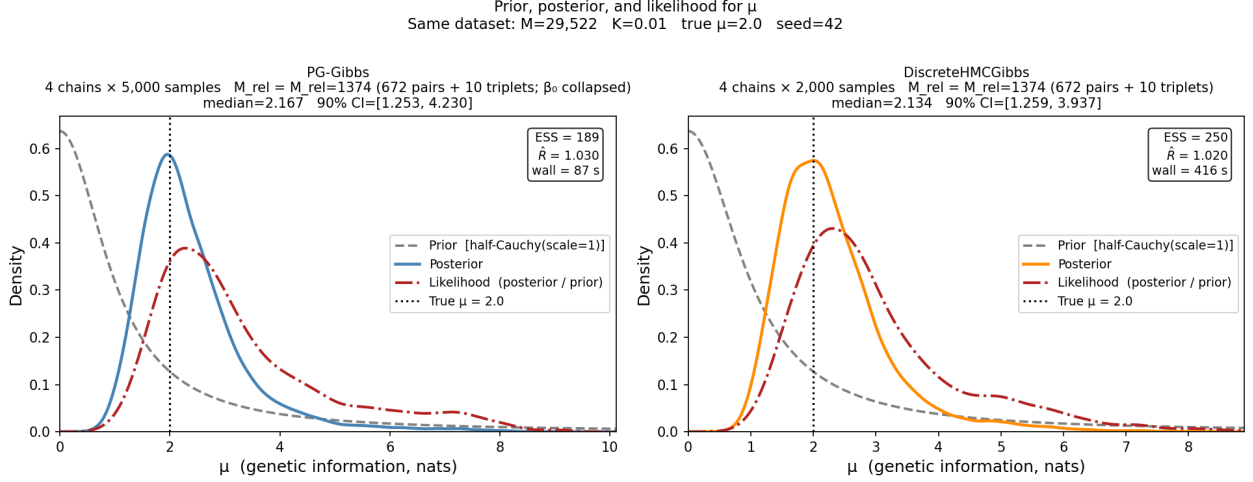


Figure 1: Prior, posterior and likelihood for both algorithms on the same dataset.

Singleton exclusion. Only the M_{rel} relatives contribute to the collapsed μ log-posterior. Including singletons (whose Z_{mix} terms do not covary with any other individual) would introduce a large spurious downward bias via the $-\frac{1}{4}M\mu$ term.

Mixing. The PG-Gibbs sampler mixes more slowly than NUTS because the μ update conditions on ω (which is sampled at the current μ), introducing autocorrelation. The collapsed update (integrating out Z_{mix} analytically) reduces but does not eliminate this dependence.

5 References

- Clayton DG (2009). Prediction and interaction in complex disease genetics: experience in type 1 diabetes. *PLoS Genetics*, 5(7), e1000540.
- McKeigue PM (2019). Quantifying performance of a diagnostic test as the expected information for discrimination: relation to the C-statistic. *Statistical Methods in Medical Research*, 28(6), 1841–1851.
- Polson NG, Scott JG, Windle J (2013). Bayesian inference for logistic models using Pólya-gamma latent variables. *Journal of the American Statistical Association*, 108(504), 1339–1349.